

TSHELO — BANK OF BOTSWANA REGULATORY SANDBOX

Comprehensive Testing Program Document

Data Sentinels | Submitted: 31 March 2026

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This document constitutes the full testing program submission for Tshelo's application to the Bank of Botswana Regulatory Sandbox. It covers the Test Plan, KPIs, Control Boundaries, Customer Communication Plan, Risk Management Framework, ICT & Cybersecurity Framework, Business Continuity Plan, AML/CFT Framework, Consumer Protection Framework, Complaints Handling Procedure, and Exit Plan. It is submitted alongside the main sandbox application form.

SECTION A — TESTING DESIGN & TEST PLAN

A1. Test Objectives

The primary objective of sandbox testing is to validate that Tshelo's transparency and record-keeping platform operates as intended in a live environment with real Botswana users, and to confirm that it does not inadvertently engage in regulated financial activity. Secondary objectives include:

- Validate SMS payment detection accuracy across Orange Money, MyZaka, and Smega
- Validate receipt scanning and expense logging under real-world conditions
- Assess user adoption, trust, and dispute resolution behaviours in live community groups
- Confirm that fraud detection logic (fund growth monitoring, fund limits) functions as designed
- Establish that all funds remain with users' own mobile money providers at all times
- Gather data to inform Version 2.0 feature development (AI expense suggestions, multi-currency)

A2. Regulatory Hypotheses

The following regulatory questions will be tested and answered during the sandbox period:

- H1: Tshelo does not constitute a payment processor or money transfer service under any applicable Botswana regulation.
- H2: Tshelo's SMS detection feature does not create a financial relationship between Tshelo and mobile money providers.
- H3: Tshelo's record-keeping function does not trigger deposit-taking obligations under the Banking Act 2023.
- H4: The platform's fraud detection mechanisms are sufficient to deter misuse given that no funds are held.

A3. Test Scenarios

Scenario	Fund Type	Description
Scenario 1	Funeral Fund	A family of 15 creates a funeral fund. Organiser uses Orange Money. Contributors u
Scenario 2	Wedding Fund	A group of 20 creates a wedding fund with a higher contribution goal. Tests fund sca
Scenario 3	Graduation Celebration	A smaller group of 8. Tests reminder functionality, dispute flagging, and audit trail int
Scenario 4	Attempted Misuse	A controlled test of fraud detection — a fund is created and contributions grow unusu

SECTION B — PERFORMANCE MONITORING & KPIs

B1. Key Performance Indicators and Targets

KPI	Target	Measurement Method
SMS detection accuracy	≥ 90% of pasted SMSs correctly parsed	System logs
User registration completion rate	≥ 80% of invited users complete OTAP registration	App analytics
Fund creation success rate	≥ 95% of attempted fund creations successful	App analytics
Contribution recording accuracy	100% of confirmed contributions correctly recorded	Activity log review
Receipt scan success rate	≥ 85% of receipt photos successfully scanned	System logs
PDF report generation success	100% of closed funds generate complete PDF reports	System PDF reports
Dispute rate	< 5% of funds experience a contribution dispute	Support records
Fraud detection trigger rate	0 funds exceed defined limits without triggering alert	System logs
User satisfaction score	≥ 4/5 average rating from post-test survey	Survey
Active fund completion rate	≥ 70% of created funds are formally closed	App analytics
Number of live funds tested	Minimum 10 live funds across 4 scenarios	App analytics
Number of users tested	Minimum 50 unique users	App analytics

B2. Success Criteria

The sandbox test will be deemed successful if: (1) all four test scenarios are completed without regulatory incident; (2) at least 8 of the 13 KPI targets above are met; (3) no user funds are lost or inaccessible as a result of Tshelo's operation; and (4) the Bank of Botswana confirms that no regulated activity has been conducted.

SECTION C — CONTROL BOUNDARIES OF THE TEST

C1. Testing Period and Milestones

- **Month 1 (April 2026):** Sandbox approval confirmed, environment setup, initial user recruitment
- **Month 2 (May 2026):** Scenarios 1 & 2 launched with controlled user groups
- **Month 3 (June 2026):** Scenarios 3 & 4 launched, mid-point review with BoB
- **Month 4 (July 2026):** All scenarios running, data collection and monitoring
- **Month 5 (August 2026):** Test wind-down, fund closures, report generation
- **Month 6 (September 2026):** Final report submitted to Bank of Botswana

C2. Customer Type Limits

Testing will be limited to retail individual users only — specifically Botswana residents who are existing mobile money users (Orange Money, MyZaka, or Smega). No institutional or corporate users will be onboarded during the sandbox period. All users must be adults (18+) who provide informed consent.

C3. Number of Customers

The sandbox test will be limited to a maximum of 200 unique registered users across all test scenarios. Minimum target is 50 users. Each fund will have a maximum of 30 members during the sandbox period.

C4. Transaction Size, Frequency, and Exposure Limits

- Maximum fund goal: BWP 10,000 per fund during sandbox testing
- Maximum individual contribution recorded per transaction: BWP 2,000
- Maximum number of active funds per user: 2 during sandbox period
- No limits on frequency of contribution recording, as Tshelo does not process transactions
- Tshelo holds zero funds at any time — financial exposure to Tshelo is nil

C5. Maximum Potential Loss

Because Tshelo does not hold, process, or transmit any funds, direct financial loss attributable to Tshelo is not possible. All funds remain in users' own mobile money accounts at all times. The maximum potential loss to users from Tshelo's operation is limited to: (1) data inaccuracy in contribution records, which is mitigated by the full audit trail and dispute resolution process; and (2) loss of access to contribution history in the event of a platform outage, mitigated by the PDF export feature. There are no indirect market effects given the limited scale of the test.

SECTION D — CUSTOMER COMMUNICATION PLAN

D1. Communication Plan Overview

All sandbox test participants will be recruited through existing community networks. Prior to onboarding, each user will receive a plain-language information pack (via WhatsApp and SMS) explaining: what Tshelo is, what the sandbox test involves, what data is collected, their rights, and how to raise a complaint. OTP verification serves as the confirmation of informed consent.

D2. Risk Disclosure Template

--- TSHELO SANDBOX TEST — USER RISK DISCLOSURE ---

Welcome to Tshelo. You have been invited to participate in a live test of the Tshelo app, which is being conducted under the supervision of the Bank of Botswana Regulatory Sandbox.

What Tshelo does: Tshelo helps your group track contributions and expenses for your event. It records who has contributed what and documents how money is spent.

What Tshelo does NOT do: Tshelo never touches your money. All payments go directly to the fund organiser's mobile money account. Tshelo only records what the organiser confirms.

Your data: Tshelo stores your name, phone number, and contribution records. You may request deletion of your data at any time by emailing hello@tshelo.com.

Risks: As this is a test, you may experience bugs or service interruptions. Your financial risk is nil — Tshelo holds no money. Record-keeping errors can be disputed and corrected.

Complaints: Contact us at hello@tshelo.com or call [number]. You may also contact the Bank of Botswana if you have unresolved concerns.

By continuing to use Tshelo, you confirm you have read and understood this disclosure.

SECTION E — RISK MANAGEMENT FRAMEWORK

E1. Risk Register

Risk	Likelihood	Impact	Mitigation
SMS parsing inaccuracy leads to wrong contribution recorded	Medium	Medium	User confirmation required before recording; full audit trail; dispute mechanism
Fund organiser misappropriates contributions	Low	High	Tshelo holds no funds; full transparency to all members; audit trail of all edits
Platform used to launder money via false fund records	Low	High	Fund limits; growth rate monitoring; Section 40 FIA reporting obligation observed
Data breach exposing user names and phone numbers	Low	Medium	Supabase RLS policies; OTP auth; data minimisation; encrypted storage
Platform outage during active fund recording	Medium	Low	PDF export available at any time; Supabase uptime SLA; manual recording
User disputes contribution record	Medium	Low	Full audit trail; in-app dispute flagging; 24-48hr support response
Regulatory classification changes lowering sandbox	Low	High	Ongoing BoB engagement; sandbox participation provides early warning

SECTION E — ICT & CYBERSECURITY FRAMEWORK

E2. ICT and Cybersecurity Controls

Authentication & Access Control:

- OTP-based authentication for all user accounts via Supabase Auth
- Phone number as primary identifier — no anonymous accounts
- Row Level Security (RLS) enforced at database level — users can only access their own fund data
- Admin access restricted to Data Sentinels engineering team with multi-factor authentication

Data Protection:

- All data encrypted at rest and in transit via Supabase (AES-256 / TLS 1.2+)
- Data minimisation — only name, phone number, and contribution records stored
- No financial account credentials stored by Tshelo at any time
- User data deletion available on request within 30 days

Infrastructure:

- Hosted on Supabase cloud infrastructure with 99.9% uptime SLA
- Automatic database backups every 24 hours
- React Native app distributed via Google Play Store and Apple App Store
- API rate limiting to prevent abuse

E3. Business Continuity Plan

In the event of a platform outage or test failure, the following measures protect users:

- All users are advised to export their PDF audit report at regular intervals during active funds
- Contribution data is backed up every 24 hours — maximum data loss in an outage is 24 hours of records
- Users retain their mobile money transaction history independently of Tshelo
- In the event of permanent platform discontinuation, all users will be notified by SMS and email with 30 days notice and provided a final PDF export of all their fund data
- As Tshelo holds no funds, no financial compensation mechanism is required — users' money is always in their own mobile money accounts

SECTION E — AML/CFT/CPF RISK MANAGEMENT FRAMEWORK

E4. AML/CFT Policy

Although Tshelo is not a specified party under Schedule I of the Financial Intelligence Act No. 2 of 2022, Data Sentinels acknowledges the general reporting obligation under Section 40 of that Act and is committed to the following:

- Tshelo will monitor fund growth patterns and flag any fund that grows at an unusual speed or exceeds defined size thresholds for internal review
- Fund size limits are enforced — initial funds capped at BWP 10,000, scaling only on the basis of completed fund history
- In the event that platform activity gives rise to a reasonable suspicion of money laundering or terrorism financing, Data Sentinels will report to the Financial Intelligence Agency as required by Section 40
- No anonymous accounts — all users verified via OTP linked to a registered mobile number
- All fund contribution records are permanent and immutable — no record can be deleted, only disputed and corrected with a visible audit trail
- Data Sentinels will appoint a designated AML compliance contact (CEO) responsible for monitoring and reporting obligations during the sandbox period

SECTION E — CONSUMER PROTECTION FRAMEWORK

E5. Consumer Protection Principles

- Full transparency: every fund member sees all contributions and expenses in real time
- Informed consent: all users receive risk disclosure before onboarding
- Data rights: users may export, correct, or request deletion of their data at any time
- No hidden fees: free tier is genuinely free; paid features are clearly disclosed before purchase
- No financial risk from Tshelo: Tshelo holds no user funds; financial risk sits with licensed mobile money providers who have their own consumer protection obligations
- Vulnerable users: Tshelo does not extend credit, accept deposits, or create financial obligations — the product is inherently low-risk for vulnerable users

E6. Customer Complaints Handling Procedure

Step 1 — In-app: User flags dispute or raises issue via in-app help centre.

Step 2 — Email: hello@tshelo.com — response within 24-48 hours.

Step 3 — Escalation: Unresolved complaints escalated to CEO (Nono Bokete) within 5 business days.

Step 4 — External: If unresolved, user is directed to the Bank of Botswana and/or NBFIRA.

Payment disputes: Directed to the relevant mobile money provider (Orange Money, MyZaka, Smega) as Tshelo holds no funds.

Record-keeping: All complaints logged with date, nature, resolution, and response time. Complaint log provided to BoB in monthly sandbox reports.

SECTION F — REPORTING AND FEEDBACK

F1. Reporting Schedule

Report	Frequency	Content	Recipient
Progress Report	Monthly	KPI performance, active funds, user numbers, complaints log	Bank of Botswana, Bank of Botswana, Bank of Botswana
Incident Report	As needed (within 48 hrs)	Any security incident, regulatory concern, data breach, etc.	Bank of Botswana, Bank of Botswana, Bank of Botswana
Mid-point Review	Month 3	Full review of all scenarios, KPI assessment, regulatory compliance status	Bank of Botswana, Bank of Botswana, Bank of Botswana
Final Report	Month 6	Complete test results, KPI achievement, regulatory findings, lessons learned, go-forth	Bank of Botswana, Bank of Botswana, Bank of Botswana

SECTION G — EXIT AND RESOLUTION PLAN

G1. Planned Exit (Successful Test)

Upon successful completion of the sandbox test, Data Sentinels will submit a final report to the Bank of Botswana and apply for approval to launch Tshelo commercially in Botswana. All sandbox test users will be notified of the successful outcome and invited to continue using the app under the commercial launch.

G2. Exit Due to Test Failure or Discontinuation

In the event that the test fails or must be discontinued, the following steps will be taken:

- All active users notified immediately via SMS and in-app notification
- All users provided with a final PDF export of their complete fund data within 7 days
- Platform remains in read-only mode for 30 days to allow users to retrieve their records
- All user data deleted from Tshelo servers within 90 days, with confirmation provided to users
- No financial compensation is required or applicable — Tshelo holds no user funds at any time. Users' money remains in their own mobile money accounts and is entirely unaffected by Tshelo's discontinuation
- A full incident report submitted to the Bank of Botswana within 14 days of discontinuation

G3. Investor Protection

Tshelo is currently self-funded by Data Sentinels. There are no external investors at this stage. In the event of test failure, Data Sentinels absorbs all development and operational costs. No user or investor funds are at risk from Tshelo's operation.

SECTION H — TECHNICAL ARCHITECTURE (Question 22)

H1. Technology Stack

Layer	Technology	Purpose
Mobile App	React Native (Expo)	Cross-platform iOS and Android application
Backend / Database	Supabase (PostgreSQL)	User data, fund records, contribution tracking, RLS security
Authentication	Supabase Auth (OTP)	Phone number + OTP verification for all users
File Storage	Supabase Storage	Receipt images and PDF report storage
SMS Detection	React Native SMS Reader (Android)	Receiving mobile money SMS notifications with user permission
Receipt Scanning	AI OCR (planned: Google Vision API)	Extracting vendor, amount, and category from receipt photos
PDF Generation	React Native PDF Library	Generating audit report PDFs at fund close
Push Notifications	Expo Push Notifications	Contribution reminders and fund updates
Hosting	Supabase Cloud (AWS-backed)	99.9% uptime SLA, AES-256 encryption at rest

H2. System Components

- User Module: Registration, OTP verification, profile management
- Fund Module: Fund creation, goal setting, member invitation, fund closure
- Contribution Module: SMS paste and parse, manual entry, contribution confirmation, audit trail
- Expense Module: Receipt photo capture, OCR extraction, expense logging, balance tracking
- Reporting Module: Real-time dashboard, PDF audit report generation, WhatsApp/SMS sharing
- Notification Module: Contribution reminders, fund updates, system alerts
- Admin Module: Fraud monitoring, fund limit enforcement, support ticket management

H3. External Integrations

- Orange Money, MyZaka, Smega: SMS notification reading (device-level, user-permissioned) — API integration planned for Version 2
- BTC API: In active discussions for direct mobile money notification integration
- Google Vision API (planned): Receipt OCR for expense scanning
- WhatsApp / SMS: Fund invite sharing via native device share functionality
- App Stores: Google Play Store and Apple App Store for distribution

H4. Data Flow Summary

User installs app → registers via phone + OTP → creates fund → shares invite link → contributors send money via their own mobile money apps directly to organiser → organiser receives SMS → pastes into Tshelo → Tshelo parses and records → organiser confirms → all members see update → organiser scans receipts → expenses logged → fund closed → PDF report generated. At no

point does any financial data or funds pass through Tshelo's servers.

— End of Tshelo Sandbox Testing Program Document | Data Sentinels | 31 March 2026 —